COMPONENTS OF ROBOTS AS A SYSTEM

Focus: Characteristics and Principles of Gears in Robotics

Use of Gears in Robots



Robotic Joints

Allows for smooth and accurate movement. Utilizes spur and harmonic drive gears.



Robot Mobility

Essential for gearbox systems powering robot wheels or tracks for varied terrain.



Lifting Power

Used in mechanisms to magnify force, allowing robots to carry heavy weights.



Accuracy

Ensures precise control and positioning in industrial automation and surgery.

Transmission of Force

The term "force transmission" in robotics describes how mechanical power gets transmitted from one part to another via the interlocking of gear teeth.

When connecting different robotic components, gears are frequently employed to make sure that force and motion generated in one part are effectively transmitted to another.



Amplification of Force

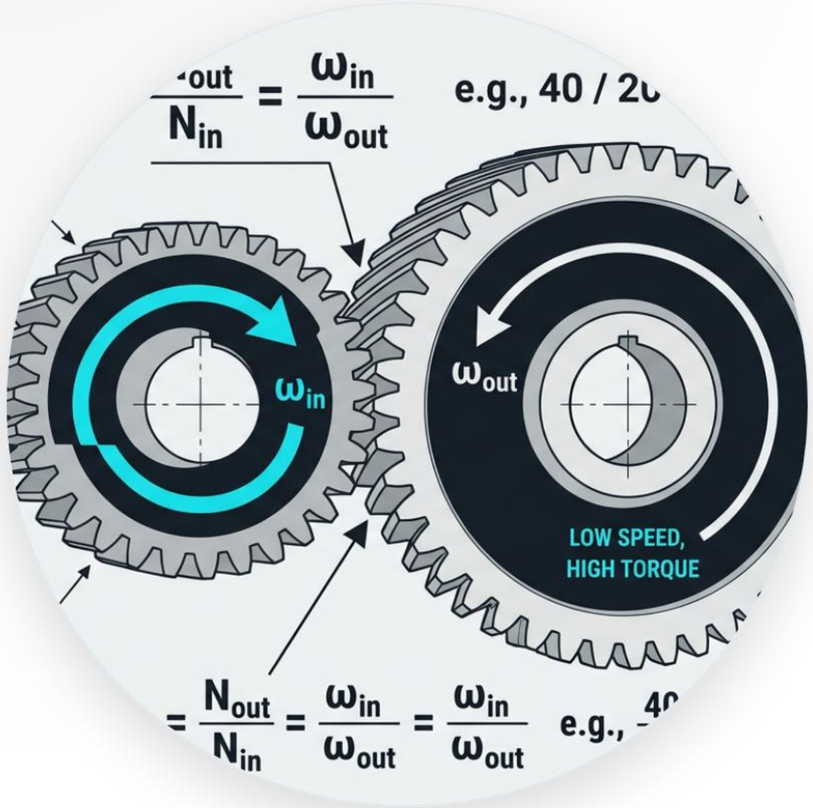


Gears increase force through gear ratios, which differ in the number of teeth on the driver gear (input) and the driven gear (output).

Engineers choose specific gear ratios to:

- Improve **Torque** (rotational force)
- Increase **Rotational Speed**

Principles of Gear Operation



Mechanical Engagement

The primary function of gears is to transmit motion and power from one component or system to another.

Gears are made up of teethed wheels that are a certain size and shape. As these wheels turn, their teeth lock together, moving power from the driver gear to the driven gear.

$$\text{Torque}_{out} = \text{Torque}_{in} \times \frac{N_{driven}}{N_{driver}}$$

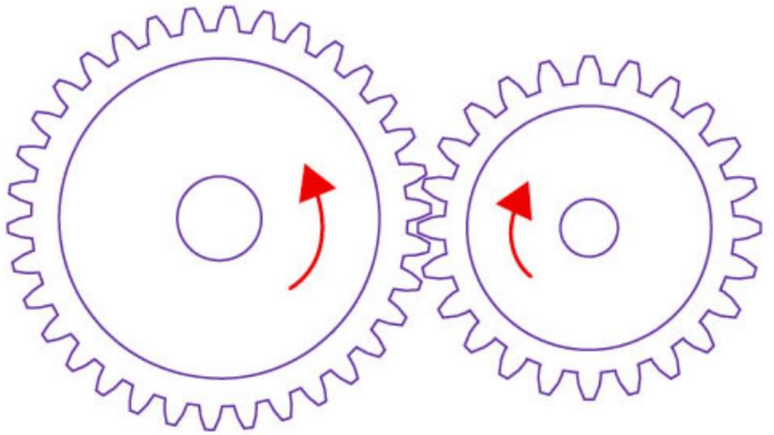
Types of Gears

A classification of common gear designs in robotics

Spur Gears

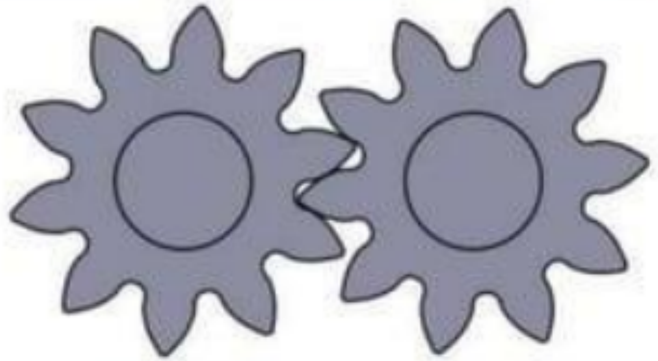
Spur gears are the most common kind of gears. They have straight teeth and are placed on shafts that are parallel to each other.

- Simplicity in design and manufacturing.
- Highly efficient for power transmission.
- Used widely for general motion in robotics.



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Helical Gears



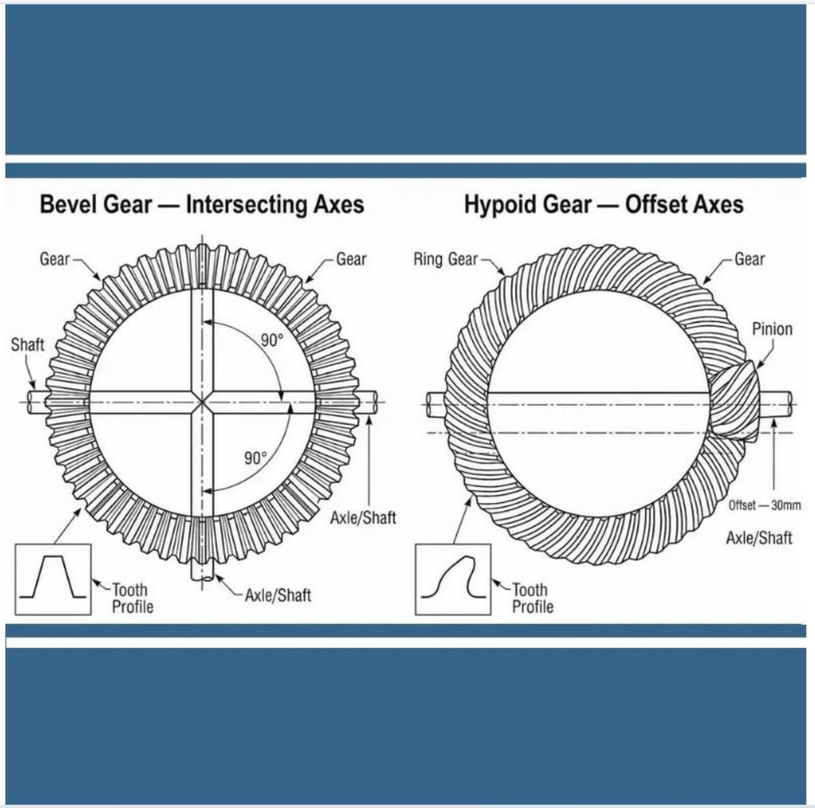
Helical gears are like spur gears, but their teeth are curved. This design choice makes them run more smoothly and quietly.

- Gradual tooth engagement.
- Higher load-carrying capacity.
- Preferred for high-speed robotic applications.

Bevel Gears

Bevel gears are used to alter the direction of movement between two shafts. Their teeth are cut on conical surfaces.

- Allows shafts to intersect at an angle (usually 90°).
- Commonly found in robot steering and differential systems.



Bevel Gears

Zero Bevel Gears

Zero Bevel gears are similar to straight bevel gears, but their teeth are curved lengthwise. These curved teeth of zero bevel gears are arranged in a manner that the effective spiral angle is zero.



Spiral Bevel Gears

- In these Spiral Bevel gears, the teeth are oblique. Spiral Bevel gears are quieter and can take up more load as compared to straight bevel gears.



Worm Gears



A worm gear is made up of a rod with threads (a "worm") and a wheel with teeth (a "worm gear").

- Enables very high gear reduction ratios.
- Provides "self-locking" features.
- Ideal for applications needing fine positional control.

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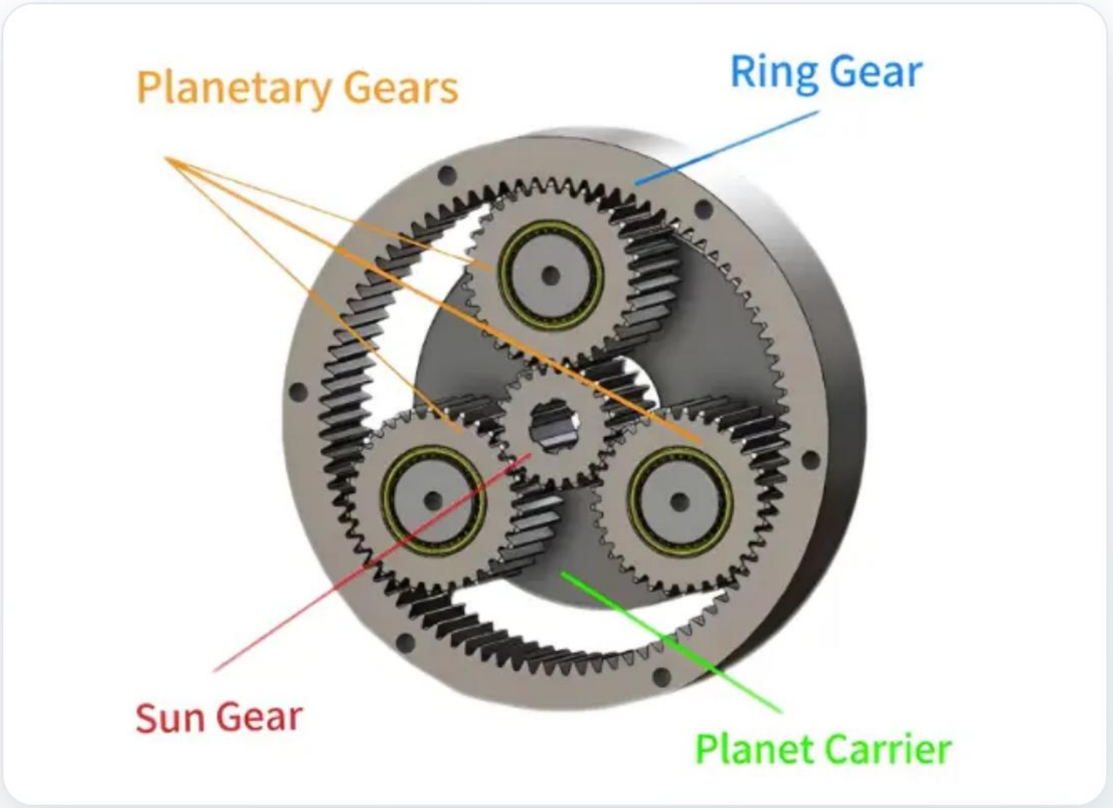
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Planetary Gears

Planetary gears consist of a central "sun" gear, multiple "planet" gears, and an outer "ring" gear.

- Frequently used in robotic arms and small engines.
- Transfer power exceptionally well in a compact space.
- High torque density.



Summary & Questions

Gears are the muscle and precision of robotic systems.

Core Takeaways:

Force Transmission | Torque Amplification | Directional Control

Image Sources



https://static.vecteezy.com/system/resources/thumbnails/070/627/070/small/close-up-of-intricate-interlocking-metal-gears-and-shafts-showcasing-precision-engineering-and-mechanical-complexity-industrial-background-photo.jpeg

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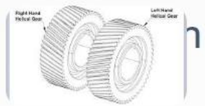
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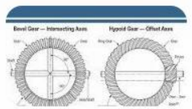
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